# Quantitative Remainder Transport in Collatz Dynamics

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**Abstract**

Let $\mathrm{Col}(n) = n/2$ for even $n$ and $\mathrm{Col}(n) = 3n+1$ for odd $n$. We combine an explicit positive-density theorem for Collatz convergence in logarithmic time with effective residue transport in inverse Collatz trees. For every modulus $N \geq 1$, every residue $r \pmod N$, and every sufficiently large parameter $X$, we construct a finite set $W_{N,r}(X)$ of distinct integers such that every $n \in W_{N,r}(X)$ satisfies

$$n \equiv r \pmod N, \qquad n \rightsquigarrow 1,$$

and

$$|W_{N,r}(X)| \geq \frac{\delta X}{2(\lceil (523/50)\log X \rceil + 4)}.$$

Here $\delta > 0$ and the cutoff are the explicit constants in Mazur's positive-density formalization. The general construction gives

$$n \leq (3X+1)2^{2^{48N^4}},$$

and we prove substantially sharper bounds when $N$ is coprime to 6, when $N$ is a power of 3, and when a factorization certificate for $N$ and the target residue is supplied. All principal statements are formalized in Lean 4.30.0-rc2 and their axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`. The result does not prove the Collatz conjecture or positive density inside each residue class; it isolates orbit coalescence as the source of the logarithmic loss.

## 1 Introduction

The Collatz map is the function $\mathrm{Col} : \mathbb{N}_{>0} \to \mathbb{N}_{>0}$ defined by

$$\mathrm{Col}(n) = \begin{cases} n/2, & 2 \mid n, \\ 3n+1, & 2 \nmid n. \end{cases}$$

The Collatz conjecture asserts that every positive integer eventually reaches 1. Although the rule is elementary, its forward and inverse dynamics combine deterministic arithmetic with behavior that repeatedly resembles a mixing process. This resemblance is useful but dangerous: qualitative coverage of residue classes is much weaker than equidistribution, and neither implies universal convergence.

This paper studies a precise interface between two rigorous phenomena. The first is abundance: a fixed explicit positive proportion of starting values reaches 1 within $(523/50)\log n$ ordinary Collatz steps. The second is transport: for a target not divisible by 3, a predecessor can be found in every prescribed residue class, with an explicit upper bound. Applying transport separately to many fast-convergent targets does not automatically produce equally many distinct predecessors, because

several targets may lie on one forward orbit. We prove that the multiplicity is at most logarithmic, which yields an explicit $X/\log X$ family in every residue class.

The central conclusion is therefore stronger than the qualitative assertion that every residue occurs infinitely often in an inverse tree. It gives a finite, explicit, growing population of integers in each residue class that are certified to reach 1.

## 2 Prior work

Terras proved fundamental density results for finite stopping time, with related work by Everett. Wirsching developed a systematic analysis of the inverse Collatz graph and later formulated the positive-predecessor-density problem. Tao proved that almost all Collatz orbits attain any prescribed diverging bound, in logarithmic density, by an approximate transport argument for the Syracuse map.

Mazur subsequently gave an explicit positive-density theorem in logarithmic ordinary time: there are explicit constants $\delta > 0$ and $X_0$ such that, for every $X \geq X_0$, at least $\delta X$ positive integers below $X$ reach 1 within $(523/50)\log n$ steps. A separate predecessor-density result shows positive lower density for the predecessors of each fixed target not divisible by 3.

Hallqvist's Every Remainder Everywhere theorem proves that the inverse tree of every target not divisible by 3 meets every residue class modulo every modulus infinitely often. Its effective version supplies the explicit general witness bound used here. These results address modular accessibility, not the number of distinct transported witnesses obtained from a population of targets.

Lau and AxiomProver recently made positive-density constants more explicit and proved target-specific predecessor results. Those results concern abundance for a prescribed target. The present theorem addresses a different axis: simultaneous quantitative transport into each prescribed congruence class. No claim of priority beyond the exact residue-wise statements formalized here is made.

## 3 Definitions

Write $a \rightsquigarrow b$ when an iterate of Col starting at $a$ equals $b$. For $C \geq 0$ and $X \in \mathbb{N}$, let

$$\mathcal{F}_C(X) = \{n < X : n \text{ is odd and reaches 1 in at most } C\log n \text{ steps}\}.$$

Define the live-target map

$$L(n) = \begin{cases} 3n+1, & 3 \mid n, \\ n, & 3 \nmid n. \end{cases}$$

For odd $n$, either $L(n) = n$ or $L(n) = \mathrm{Col}(n)$. Thus $L(n)$ is not divisible by 3, reaches 1, and requires no more steps to reach 1 than $n$ does. Let

$$\mathcal{T}_C(X) = L(\mathcal{F}_C(X)).$$

The map $L$ has fibres of size at most two on odd inputs, hence

$$|\mathcal{F}_C(X)| \leq 2|\mathcal{T}_C(X)|. \tag{1}$$

## 4 The orbit-fibre bound

Suppose a selector $s$ assigns to each $w \in \mathcal{T}_C(X)$ a predecessor $s(w)$ with $s(w) \rightsquigarrow w$. If $s(u) = s(v)$, then both $u$ and $v$ occur on the same forward orbit, namely the orbit of the common selected predecessor. If every target reaches 1 within $B$ steps, all such targets lie among a prefix of length $B+1$ together with the terminal cycle $1, 4, 2$. Consequently each fibre of $s$ has cardinality at most $B+4$.

**Proposition 4.1** (Selected-witness counting)**.** *If every $w$ in a finite target set $S$ reaches 1 within $B$ steps and $s(w) \rightsquigarrow w$, then*

$$|S| \le |s(S)|(B+4).$$

For $C = 523/50$, one may take

$$B = B(X) = \left\lceil \frac{523}{50} \log X \right\rceil.$$

Combining the proposition with (1) gives

$$|\mathcal{F}_{523/50}(X)| \le 2|s(\mathcal{T}_{523/50}(X))|(B(X)+4). \tag{2}$$

## 5 Effective residue transport

Effective Remainder Everywhere supplies, for every $N \ge 1$, residue $r$, and live target $w$, a selected predecessor $s_{N,r}(w)$ such that

$$s_{N,r}(w) \rightsquigarrow w, \qquad s_{N,r}(w) \equiv r \pmod N,$$

with

$$s_{N,r}(w) \le w 2^{2^{48N^4}}. \tag{3}$$

Since every target in $\mathcal{T}_C(X)$ is below $3X+1$, the right-hand side of (3) is at most $(3X+1)2^{2^{48N^4}}$.

## 6 Main theorems

Let $\delta > 0$ and $X_0$ be Mazur's explicit density constant and cutoff.

**Theorem 6.1** (Quantitative direct-to-one remainder transport)**.** *For every $N \ge 1$, $r \in \mathbb{N}$, and $X \ge X_0$, there is a finite set $W_{N,r}(X)$ satisfying*

$$|W_{N,r}(X)| \ge \frac{\delta X}{2(B(X)+4)}.$$

*Every $m \in W_{N,r}(X)$ satisfies*

$$m \equiv r \pmod N, \qquad m \rightsquigarrow 1, \qquad m \le (3X+1)2^{2^{48N^4}}.$$

*Proof.* Mazur's theorem gives $\delta X \le |\mathcal{F}_{523/50}(X)|$. Apply the effective residue selector to every live target and then use (2). The selected image is a finite set of distinct integers in the prescribed residue. Each selected integer reaches its target and each target reaches 1, so transitivity gives convergence to 1. The numerical bound follows from (3) and $w < 3X+1$. □

**Theorem 6.2** (Coprime-to-six refinement)**.** *If* $\gcd(N,6)=1$*, the same cardinality conclusion holds with*

$$m \le (3X+1)2^{2^{15(3N^2)}} = (3X+1)2^{2^{45N^2}}.$$

**Theorem 6.3** (Power-of-three refinement)**.** *For* $N=3^b$*, the same cardinality conclusion holds with*

$$m \le (3X+1)2^{2^{15(3\cdot 3^b)}} = (3X+1)2^{2^{45N}}.$$

**Theorem 6.4** (Factor-sensitive refinement)**.** *Let* $N=2^\alpha 3^b M$ *with* $\gcd(M,6)=1$*, and suppose a certified parity word and auxiliary integers satisfy the hypotheses of the formal factorized transport theorem. If* $Z$ *is the resulting effective scale, then the same cardinality conclusion holds with*

$$m \le 2N(3X+1)2^{2^{15Z}}.$$

*The complete arithmetic certificate is stated in the Lean declaration* `direct_to_one_logarithmic_count_factorized`

# 7 What the theorem means for Collatz dynamics

The theorem gives a quantitative form of modular permeability. No fixed congruence class can be only sparsely accessible in the qualitative sense of having a few exceptional predecessors: every class receives an explicitly growing family of certified convergent integers.

This should not be confused with uniform randomness. The result neither compares different residue classes nor proves that their counts have an asymptotic ratio of one. The construction may be highly biased and its numerical bounds are enormous. What it rules out is a broad family of purely modular obstructions to convergence-to-one membership.

The logarithmic denominator has a dynamical interpretation. Effective residue transport itself does not cause the loss. The loss arises because different fast targets can lie on the same orbit and therefore select the same predecessor. A Collatz orbit segment of logarithmic length can absorb at most logarithmically many targets. Thus the remaining step from the present $X/\log X$ guarantee to a positive-density guarantee in each residue class is a concrete multiplicity problem, not a modular reachability problem.

In a physical or statistical-mechanical analogy, the inverse dynamics has full modular support but has not been shown to have a uniform stationary distribution. Coverage is analogous to irreducibility; positive density or equidistribution would require quantitative control of mass concentration. This analogy is interpretive only and is not used in the proof.

# 8 Relation to the full Collatz conjecture

The theorem does not prove that every integer reaches 1, that almost every integer in each residue reaches 1, or even that the convergent integers have positive lower density within each residue class. Its exact finite lower bound is of order $X/\log X$ in the target parameter, while transported witnesses lie below a modulus-dependent constant multiple of $X$.

Nevertheless, the theorem sharpens the map of what a counterexample mechanism would have to accomplish. It cannot rely merely on excluding a residue class from convergent inverse trees. Any persistent nonconvergent component must coexist with quantitatively many convergent starts in every congruence class.

## 9 Further directions

The most immediate problem is to replace the worst-case fibre bound $B + 4$ by a constant or a bounded average on a positive-density subfamily. Such a result would upgrade the construction toward positive density in every residue class. Computational experiments can measure fibre histograms for different target selections and test whether collision multiplicity remains bounded, has a light tail, or grows logarithmically.

Other directions include optimizing the enormous effective transport bounds, comparing counts across residue classes, transporting stronger stopping-time distributions, and replacing the external Effective Remainder Everywhere dependency with an independent formalization.

## 10 Lean formalization

The development uses Lean 4.30.0-rc2 with a pinned Mathlib revision. The principal declarations are:

- `explicit_logarithmic_remainder_transport_ten46`;
- `direct_to_one_logarithmic_count_ten46`;
- `direct_to_one_count_of_selector`;
- `direct_to_one_logarithmic_count_coprime_six`;
- `direct_to_one_logarithmic_count_pow_three`;
- `direct_to_one_logarithmic_count_factorized`.

For each, `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound`. No `sorry` or `admit` occurs in the audited theorem modules.

The initial public release does not redistribute Hallqvist's unlicensed Lean source. It records that source as an external prerequisite by repository URL, filename, and hash. Mazur's required source closure is redistributed under its Apache-2.0 terms with its notice retained.

## Authorship and AI disclosure

The project architecture, selection of the synthesis problem, iterative verification process, theorem packaging, and publication are attributed to the named author. OpenAI language models accessed through ChatGPT/Codex contributed substantially to mathematical exploration, Lean proof generation, debugging, and manuscript drafting. The upstream Mazur development also discloses substantial generative-AI assistance, and Lau's refinement discloses autonomous AxiomProver contributions. AI systems are not listed as authors or independent referees. The named author assumes responsibility for the claims and released artifacts.

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